Demo Study E: Ensemble Learning for Rain Fade Mitigation: An Empirical Comparison

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SYNTHETIC DEMO DOCUMENT - fictional content for software testing. Not a real publication.

Abstract

We compare random forest, support vector regression, and an ANN ensemble for rain fade prediction using measurements from two Ku-band links. The ensemble improves accuracy over individual learners.

Introduction

Fade mitigation techniques need fast, accurate predictions. Ensemble methods can reduce variance. This work aims to quantify the benefit of ensembling for short-term attenuation prediction.

Data

Measurements were recorded over 12 months on two Ku-band links using a beacon receiver and a disdrometer. The dataset includes 2.1 million samples.

Methodology

A random forest, SVR and a multilayer perceptron are combined via stacking. Evaluation uses RMSE, MAPE and the coefficient of determination R2 with k-fold cross-validation.

Results

The ensemble achieves an RMSE of 0.95 dB, improving on the best single learner by 12 percent. R2 reaches 0.91. Seasonal differences were observed, with larger errors in the monsoon months.

Limitations

Cross-validation was performed randomly rather than by time or site, which may overestimate performance. Extreme rain events remain poorly predicted, and the study covers only one year.